AUTHORIZED SECURITY LAB ASSESSMENT REPORT

RabTech Academy – Cybersecurity Internship

Project Title: Legal Scope, Asset Inventory & Threat Model

Prepared By: [Your Name]

Organization: RabTech Academy

Domain: Cybersecurity – Ethical Hacking & VAPT

Assessment Type: Authorized Security Lab Assessment

Date: [Submission Date]

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1. ABSTRACT

This project focuses on preparing an authorized cybersecurity lab environment for security assessment. The main objective is to identify permitted assets, understand data flow, analyze potential security threats and establish safe testing guidelines.

The project includes an asset inventory, Data Flow Diagram, STRIDE threat model and Rules of Engagement. These components help define the testing scope, identify possible security risks and recommend suitable security controls.

The assessment is designed around an isolated local training web application and a deliberately vulnerable virtual machine. The project follows responsible security testing practices and avoids unauthorized access to external systems.

2. INTRODUCTION

Cybersecurity assessment is an important process used to identify weaknesses in computer systems and applications.

Before performing penetration testing, it is necessary to understand the systems involved, their security boundaries and possible threats.

This project prepares an authorized security lab assessment framework using asset inventory, data flow analysis, STRIDE threat modelling and Rules of Engagement.

3. PROJECT OBJECTIVES

1. Create an isolated and authorized security lab environment.
2. Identify assets and their testing boundaries.

3. Understand data flow between system components.
4. Identify possible threats using the STRIDE methodology.
5. Recommend suitable security controls.
6. Establish safe testing rules and evidence-handling procedures.

4. ASSET INVENTORY

The proposed assessment includes two assets:

LAB-01 – Local Training Web Application

- Address: 127.0.0.1:8080
- Environment: Localhost
- Purpose: Web application security assessment.

LAB-02 – Vulnerable Virtual Machine

- Address: 192.168.56.20
- Environment: Private host-only network
- Purpose: Virtual machine security assessment.

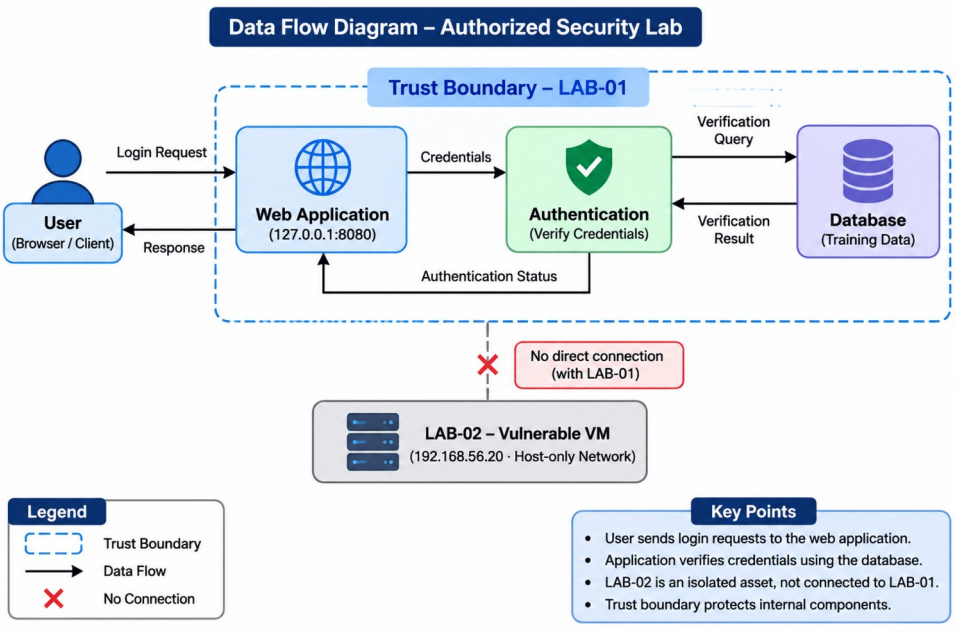
Only explicitly approved assets are included in the testing scope.

5. DATA FLOW DIAGRAM

The Data Flow Diagram explains the communication between the user, web application, authentication process and database.

Insert your Step 3 Data Flow Diagram image here.

The diagram identifies the trust boundary and shows how login requests and verification responses move between the components.



LAB-02 is represented as a separate isolated asset.

6. STRIDE THREAT MODEL

The STRIDE methodology is used to identify six major categories of security threats.

1. Spoofing – Unauthorized identity impersonation.
2. Tampering – Unauthorized modification of information.
3. Repudiation – Denying an action without sufficient audit evidence.
4. Information Disclosure – Exposure of confidential information.
5. Denial of Service – Disruption of application availability.
6. Elevation of Privilege – Obtaining unauthorized permissions.

Recommended security controls include strong authentication, input validation, audit logging, encryption, rate limiting and role-based access control.

7. RULES OF ENGAGEMENT

The Rules of Engagement define the conditions for conducting security testing.

The assessment must follow these requirements:

- Obtain written authorization.
- Test only approved assets.
- Follow the approved testing window.
- Avoid destructive activities.
- Protect collected evidence.
- Stop testing if unexpected issues occur.
- Document findings and remediation recommendations.

8. EXPECTED SECURITY BENEFITS

The proposed security assessment framework is expected to provide the following benefits:

- Better understanding of the lab environment.
- Clear identification of permitted testing assets.
- Improved awareness of possible security threats.

- Better protection of sensitive information.
- Clear testing boundaries and safety procedures.
- Organized documentation of security findings.

These are expected benefits and have not been presented as verified testing results.

9. CONCLUSION

This project establishes a structured approach for preparing an authorized cybersecurity assessment. By combining asset inventory, data flow analysis, STRIDE threat modelling and Rules of Engagement, the project provides a foundation for safe and responsible security testing.

The prepared documentation helps identify potential risks, define testing boundaries and recommend suitable security controls. Further implementation and authorized testing can be performed to validate the identified risks and recommendations.

10. REFERENCES

1. OWASP – https://owasp.org/
2. OWASP Threat Modeling – https://owasp.org/www-community/Threat_Modeling
3. Microsoft STRIDE Threat Modeling – https://learn.microsoft.com/
4. diagrams.net – https://app.diagrams.net/
5. RabTech Academy – Internship Starter Files.